| Question | Answer | Marks | Partial Marks |
|---|---|---|---|
| 3(b) | 182.875 ... 307.125 final answer | **3** | **B2** for either seen<br><br>or **M1** for $10 \pm 0.5$ or $6 \pm 0.5$ or $4 \pm 0.5$ oe |
| 4(a) | Correctly eliminate one variable | **M1** | |
| | $p = 3$<br>$q = -1$ | **A2** | **A1** for each<br>If M0, **SC1** for 2 values satisfying one of original equations<br>If 0 scored **SC1** for correct answers with no working |
| 4(b) | $1\frac{1}{11}$ or $\frac{12}{11}$ 1.09 or 1.090 to 1.091 | **2** | **M1** for $\frac{3x}{12}+\frac{8x}{12}=1$ or better |
| 4(c)(i) | $-2 < x \leqslant 3$ | **3** | **B2** for $-2 < x$ or $x \leqslant 3$<br><br>or **M1** for $-8 + 2 < 3x$ or $3x \leqslant 7 + 2$ |
| 4(c)(ii) | –1, 0, 1, 2, 3 | **1** | **FT** dep on –ve and +ve values in *their* **(c)(i)** |
| 4(d) | $4a(4-a)$ final answer | **2** | **B1** for any correct partial factorisation |
| 4(e)(i) | $\frac{2b}{3a}$ final answer | **2** | **M1** for $\frac{1}{2a}\times\frac{4b}{3}$ or better |
| 4(e)(ii) | $\frac{x-2}{x-1}$ final answer nfww | **2** | **B1** for $2(x-1)-x$ oe seen. |
| 5(a)(i) | 105 | **2** | **M1** for $\frac{3}{100}\times 500[\times 7]$ |
| 5(a)(ii) | 115 or 114.9... | **3** | **M2** for $500\times\left(1+\frac{3}{100}\right)^7[-500]$<br>or **M1** for $500\times\left(1+\frac{3}{100}\right)^k$, $k$ integer $\geqslant 2$ |
| 5(b) | 8600 | **3** | **M2** for $\frac{6269.4}{\left(1-\frac{10}{100}\right)^3}$ oe<br>or **M1** for $C\times\left(1-\frac{10}{100}\right)^3=6269.4$ oe |
| 6(a) | 9.33 or 9.334... | **3** | **M2** for $\frac{12\sin 50}{\sin 100}$<br>or **M1** for $\frac{\sin 100}{12}=\frac{\sin 50}{AD}$ oe |

| Question | Answer | Marks | Partial Marks |
|---|---|---|---|
| 6(b) | [cos =] $\frac{11^2+12^2-8^2}{2\times11\times12}$ | **M2** | **M1** for $8^2=11^2+12^2-2\times11\times12\cos(BAC)$ |
| | 40.415... | **A2** | **A1** for 0.761... or $\frac{201}{264}$ or $\frac{67}{88}$ |
| 6(c) | 70.8 or 70.77 to 70.79... | **3** | **M1** for $\frac{1}{2}\times12\times$ *their* **(a)** $\times\sin(180-100-50)$<br>**M1** for $\frac{1}{2}\times12\times11\times\sin(40.42)$ |
| 6(d) | 7.13 or 7.131 to 7.132... | **3** | **M2** for $\frac{\text{dist}}{11}=\sin(40.42)$<br>or **M1** for recognition that shortest distance is perpendicular to $AC$ |
| 7(a) | 87 | **3** | **M2** for $3c + 4c = 587 + 22$ or better<br>or **M1** for $3c + 2(2c - 11)$ [= 587 or 5.87] |
| 7(b) | 1.1[0] | **3** | **M2** for $22w + 22 = 42w$ or better<br>or **M1** for $\frac{22}{w} = \frac{42}{w+1}$ oe<br>OR<br>**B2** for number of bottles = 20<br>or **M1** for $Nw = 22$ and $N(w+1) = 42$ |
| 7(c)(i) | $\frac{9}{x}+\frac{5}{2x+1}=2.5$ oe | **M2** | **M1** for $\frac{9}{x}$ or $\frac{5}{2x+1}$ |
| | $9(2x+1)+5x=2.5x(2x+1)$ oe<br>or $\frac{9(2x+1)+5x}{x(2x+1)}$ [= 2.5 oe] | **M1** | Correctly clearing fractions, or correctly collecting into a single fraction<br>FT *their* expression dep on two fractions both with algebraic denominators |
| | All brackets expanded leading to $10x^2-41x-18=0$ with no errors or omissions | **A1** | |